



TRINITY GRAMMAR SCHOOL

Mathematics Department

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(NESA Student Number | Year 12 only)

2024

Year 12

Mathematics Advanced

TRIAL EXAMINATION

ASSESSMENT TASK 4

Date of Assessment Task: Tuesday, 27 August 2024

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Erasable pens and all electronic devices (or timekeepers), unless otherwise advised, are not permitted in this Assessment
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- *Write your NESA Student Number (Year 12) at the top of this*

Total marks:
100

Section I – 10 marks (pages 2 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7 – 30)

- Attempt Questions 11 – 31
- Allow about 2 hours and 45 minutes for this section

- **Year 12 Assessment Weighting: 30%**

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What values of x satisfy $3 - 2x \geq 8$?

A. $x \leq -\frac{11}{2}$

B. $x \geq -\frac{11}{2}$

C. $x \leq -\frac{5}{2}$

D. $x \geq \frac{5}{2}$

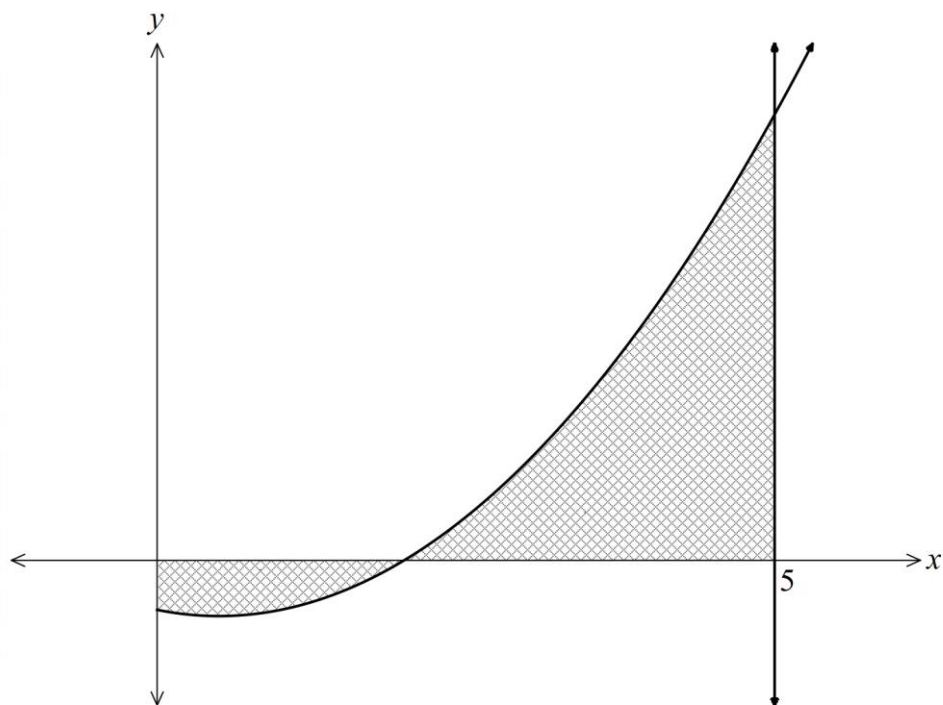
2 Which of the following represents the domain of the function $f(x) = \ln(x+1)$?

A. $[-\infty, -1)$

B. $(-\infty, 1)$

C. $[-1, \infty)$

D. $(-1, \infty)$



- 3 The shaded region in the diagram above shows the area, A , bounded by the curve $y = x^2 - x - 2$, the x -axis and the lines $x = 0$ and $x = 5$?

Which expression will give the area, A , of the shaded region?

A. $A = \left| \int_0^1 (x^2 - x - 2) dx \right| + \int_1^5 (x^2 - x - 2) dx$

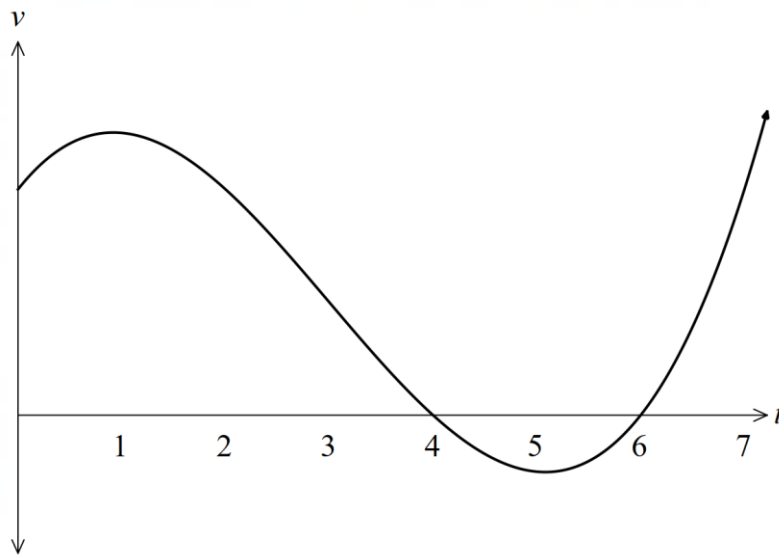
B. $A = \int_0^2 (x^2 - x - 2) dx + \left| \int_2^5 (x^2 - x - 2) dx \right|$

C. $A = \int_2^0 (x^2 - x - 2) dx + \int_2^5 (x^2 - x - 2) dx$

D. $A = \int_1^5 (x^2 - x - 2) dx$

- 4 Which of the following represent respectively the radius and coordinates of the centre of the circle $x^2 - 6x + y^2 + 8y + 21 = 0$?
- A. Radius of 2 **and** centre $(-3, 4)$
- B. Radius of 2 **and** $(3, -4)$
- C. Radius of 4 **and** $(-3, 4)$
- D. Radius of 4 **and** $(3, -4)$
- 5 What is the period of $y = \tan 3x$?
- A. $\frac{\pi}{3}$
- B. $\frac{\pi}{6}$
- C. $\frac{2\pi}{3}$
- D. 3π
- 6 The values of x for which the geometric series $2 + 6x + 18x^2 + \dots$ has a limiting sum are:
- A. $x < \frac{1}{3}$
- B. $x \leq \frac{1}{3}$
- C. $-\frac{1}{3} < x < \frac{1}{3}$
- D. $-\frac{1}{3} \leq x \leq \frac{1}{3}$

- 7 A particle is moving along a straight line. The graph below shows the velocity, v , of the particle at any time t seconds.



For what value of t is the displacement, x , a maximum?

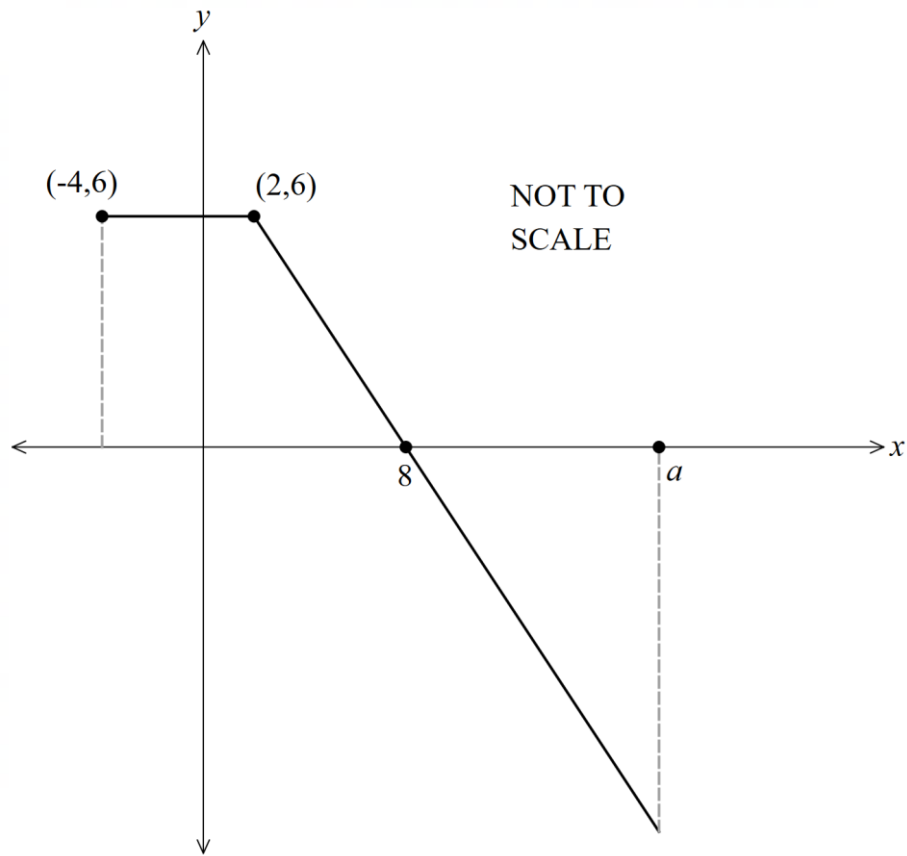
- A. 1
 B. 4
 C. 5
 D. 6
- 8 Consider the following probability distribution table, where a is a constant.

x	1	2	3	4	5
$P(X = x)$	a	$3a$	$5a$	$2a$	a

What is the mean, $E(X)$, for this probability distribution?

- A. $\frac{1}{12}$
 B. $1\frac{3}{12}$
 C. $2\frac{11}{12}$
 D. 3

- 9 The graph of $y = f(x)$ is a piecewise function with a domain of $[-4, a]$, where $a > 8$.



Find the value of a given that $\int_{-4}^a f(x) \, dx = 4$.

- A. 16
B. 18
C. 20
D. 22
- 10 How many solutions does the equation $x = 10 \sin x$ have, given $x \in [-3\pi, 3\pi]$?
- A. 4
B. 5
C. 6
D. 7



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(NESA Student Number | Year 12 only)

2024

Mathematics Advanced

Section II Answer Booklet

90 marks

Attempt Questions 11-31

Allow about 2 hours and 45 minutes for this section

Instructions

- Write your NESA Student Number (Year 12) at the top of this page
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
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Please turn over

Question 11 (4 marks)

The first three terms of an arithmetic sequence are 4, 9 and 14.

- (a) Find the 20th term. 2

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- (b) Find the sum of the first 20 terms. 2

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Question 12 (4 marks)

Find the value(s) of x such that $25^{x^2-2x} = \left(\frac{1}{125}\right)^{4x+2}$.

4

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Question 13 (3 marks)

Find the equation of the tangent to the curve $y = \frac{1}{\sqrt{x}}$ at the point $\left(4, \frac{1}{2}\right)$.

Give your final answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.

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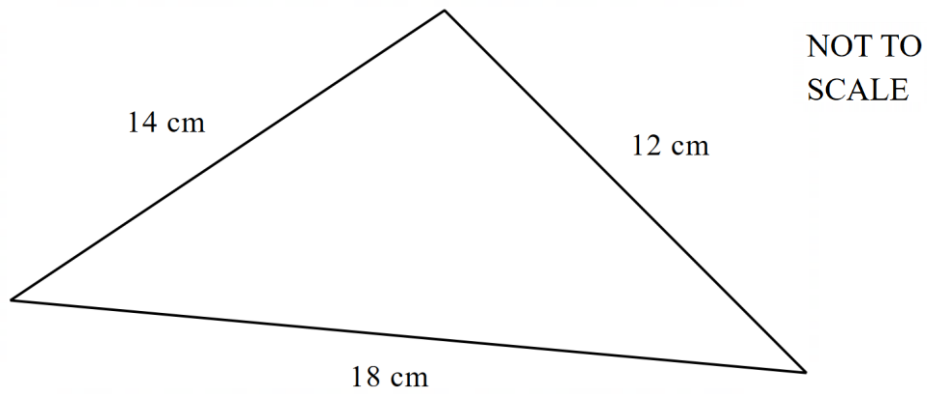
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Question 14 (3 marks)

Find the area of the triangle below. Round your answer to 1 decimal place.

3



Question 15 (2 marks)

Find $\int \frac{5x}{x^2+1} dx$.

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Question 16 (5 marks)

Differentiate each of the following expressions with respect to x , leaving your answer in simplest form where possible:

(a) $(2x^3 - 1)^5$ 1

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(b) $x \cos x$ 2

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(c) $\frac{2x}{\ln x}$ 2

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Question 17 (2 marks)

Find all the values of θ , where $0 \leq \theta \leq \pi$, such that $\sin\left(2\theta - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$ 2

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Question 18 (3 marks)

A and B are independent events with $P(A) = 0.28$ and $P(A \cup B) = 0.75$.

Find $P(B)$. 3

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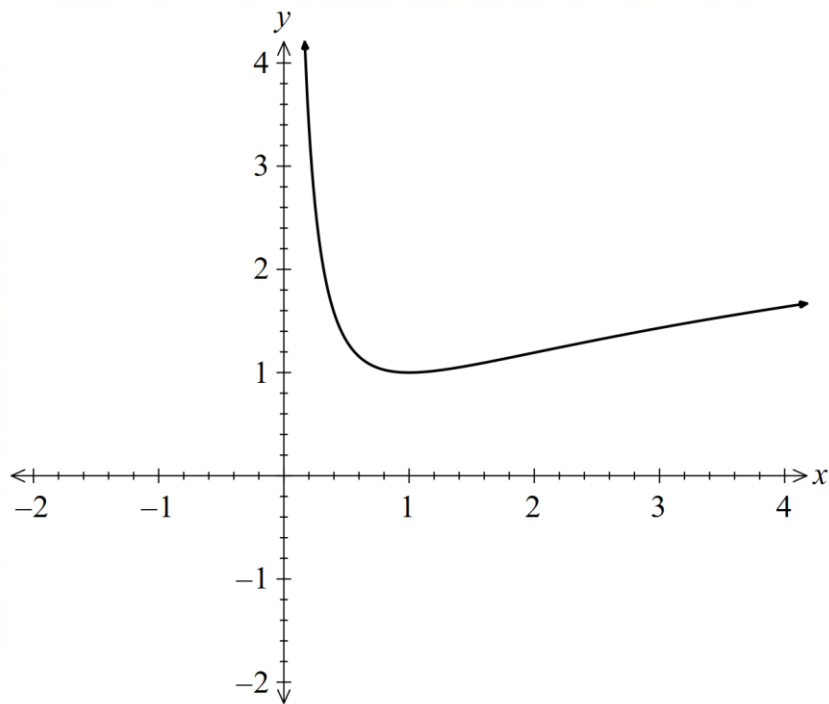
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Question 19 (2 marks)

The graph of $y = \frac{1}{x} + \ln(x)$ is shown over part of its domain.



Use **three** applications of the trapezoidal rule to find the approximate area between the curve, the x -axis and the lines $x=1$ and $x=4$. Give your answer correct to 2 decimal places.

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Question 20 (5 marks)

A table of future value interest factors for an annuity of \$1000 is shown.

<i>Period \ Rate</i>	<i>Interest rate per period</i>				
	1%	1.5%	3%	4.5%	6%
3	3030.10	3045.23	3090.90	3137.03	3183.60
6	6152.02	6229.55	6468.41	6716.89	6975.32
12	12682.50	13041.21	14192.03	15464.03	16869.94
24	26973.46	28633.52	34426.47	41689.20	50815.58

- (a) Taliah is saving for a new car. She considers contributing \$7 500 each year for the next 6 years.

If the interest rate is 6% per annum, compounding annually, how much money will Taliah have saved after 6 years?

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- (b) Taliah requires \$85 000 to buy her dream car.

She decides to make regular payments into an account earning 6% per annum, compounding **quarterly** for the next 6 years.

Given that Taliah will make her contribution every 3 months, how much more will she pay per year, than in part (a)?

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Question 21 (3 marks)

Solve the equation $2\log_5(4) - \log_5(2) + \log_5(x) = 2$ for x .

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Question 22 (3 marks)

Consider the function f such that $f'(x) = \frac{8x}{x^2 + 4}$. The graph of $y = f(x)$ passes through the point $(2, 9\ln 2)$.

Find $f(x)$.

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Question 23 (5 marks)

Let $f(x) = xe^x$ and $g(x) = 4x - x^2$.

Solve $(f \circ g)(x) = 0$.

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Please turn over

Question 24 (9 marks)

The curve $y = x^3 + ax^2 + 7x$ has a stationary point at $x = 1$.

- (a) Find the value of a . 2

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- (b) Find the coordinates of all stationary points **and** determine their nature. 3

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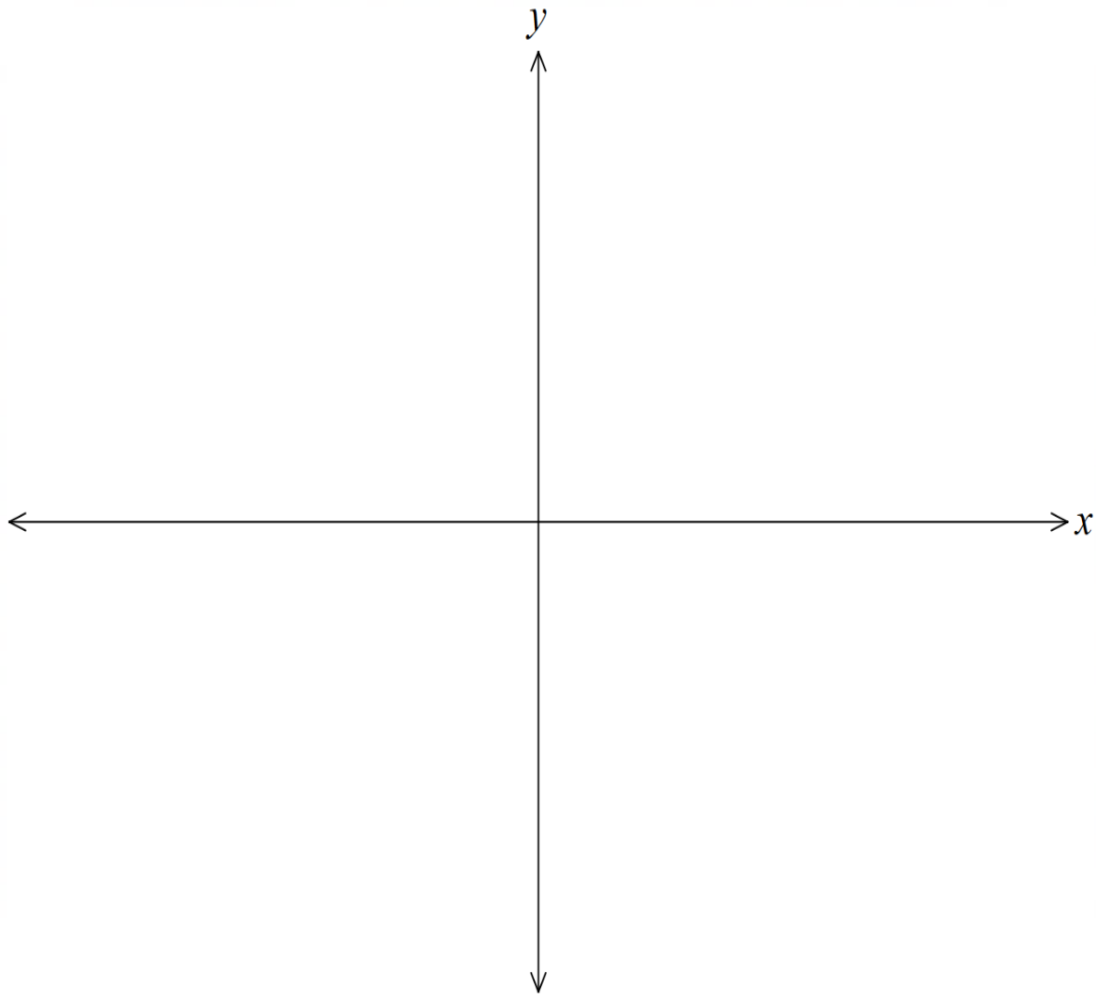
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Question 24 continues on page 19

Question 24 (continued)

- (c) Sketch the graph of $y = f(x)$ showing any stationary points and its y -intercept.

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- (d) Find any values of x such that the curve is concave up.

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End of Question 24

Please turn over

Question 25 (3 marks)

- (a) Find $\frac{d}{dx}(e^{\tan x})$. **1**

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- (b) Hence, or otherwise, find $\int \frac{2e^{\tan x}}{3-3\sin^2 x} dx$. **2**

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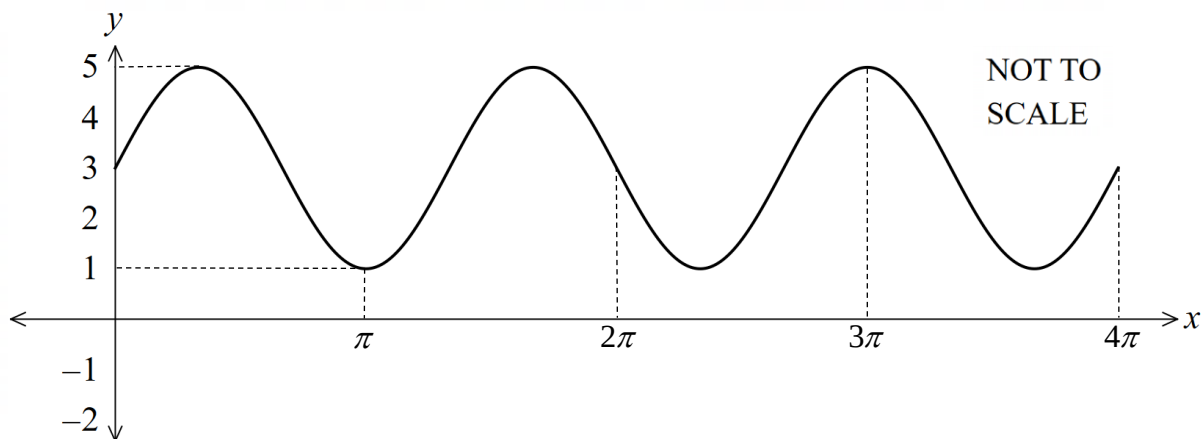
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Question 26 (4 marks)

A trigonometric function $y = f(x)$ is drawn below in the domain $[0, 4\pi]$.



- (a) State the amplitude of $y = f(x)$.

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- (b) Determine the equation of the function $y = f(x)$.

Express your final answer in the form $y = a \sin(bx) + d$.

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Question 27 (7 marks)

The following table shows values of x and y .

x	4	13	26	30	35	50
y	1.8	2.9	3.8	4.2	5.7	5.1

- (a) Calculate Pearson's correlation coefficient for the data represented in the table above, correct to 2 decimal places.

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- (b) Identify the direction **and** strength of the linear association between x and y .

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- (c) State the equation of the least-squares regression line in the form $y = ax + b$.

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- (d) Use the regression equation from part (c) to estimate the value of x when $y = 4.8$ to 2 decimal places.

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Question 28 (5 marks)

A continuous random variable X has a function given by

$$f(x) = \begin{cases} \pi \sin(2\pi x) & \text{if } 0 \leq x \leq \frac{1}{2} \\ 0 & \text{elsewhere} \end{cases}$$

- (a) Show that the function $f(x)$ is a probability density function.

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- (b) Find the mode of X .

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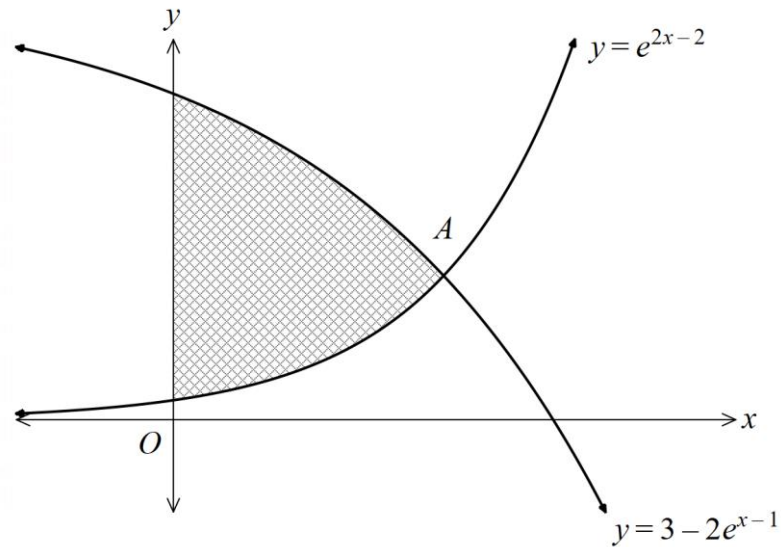
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Question 29 (4 marks)

The shaded region is bound by the functions $y = e^{2x-2}$, $y = 3 - 2e^{x-1}$ and the y -axis, as shown in the diagram. Point A is the intersection of the two functions.



- (a) Show that the x -coordinate of point A is $x = 1$.

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Question 29 continues on page 25

Question 29 (continued)

(b) Find the **exact** area of the shaded region.

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End of Question 29

Please turn over

Question 30 (8 marks)

Kareem borrows \$140 000 to start his own business. He plans to repay the loan in equal monthly instalments of \$ M at 6% per annum reducible interest. Interest is calculated and charged just before each repayment is made.

Let A_n be the amount owing after n repayments (in dollars).

- (a) Show that $A_n = 140\,000(1.005)^n - 200M(1.005^n - 1)$.

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Question 30 continues on page 27

Question 30 (continued)

- (b) Kareem would like to repay the loan in exactly 15 years. Calculate the amount of each instalment.

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- (c) Kareem can afford to repay \$1 850 per month. How many years would it take him to repay the loan?

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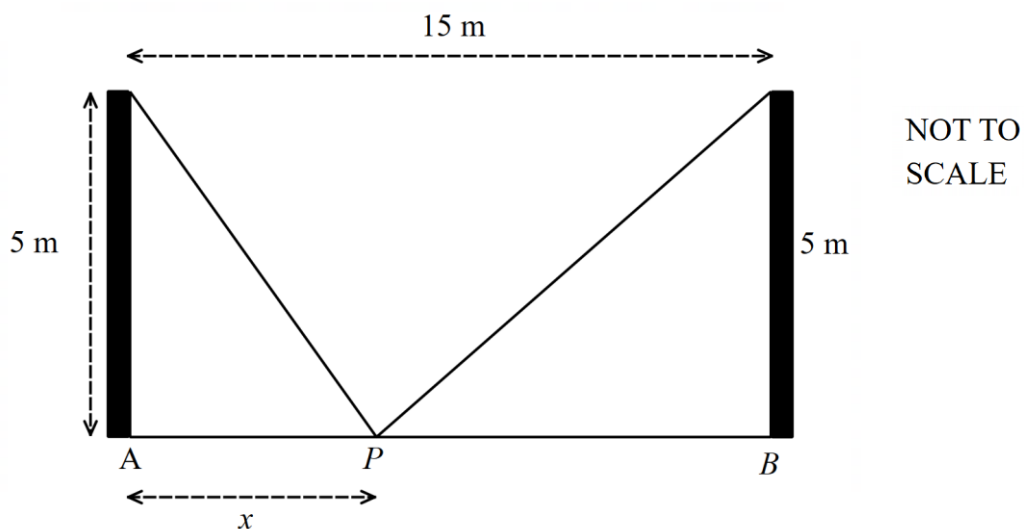
End of Question 30

Please turn over

Question 31 (6 marks)

The diagram shows two 5 m poles, A and B , that are 15 metres apart. The ends of a single length of wire are attached to the top of each pole. The length of wire is then attached to a stake in the ground between them at a moveable point, P , as shown in the diagram below.

Let x be the distance between A and P .



- (a) Show that the length of the wire, L , is given by $L = \sqrt{x^2 + 25} + \sqrt{x^2 - 30x + 250}$.

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Question 31 continues on page 29

Question 31 (continued)

- (b) Show that $\frac{dL}{dx} = \frac{x}{\sqrt{x^2 + 25}} + \frac{x-15}{\sqrt{x^2 - 30x + 250}}$. 2

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- (c) Find the value of x needed to minimise the length of wire, L used.

Justify your answer.

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End of Question 31
End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering

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Do NOT write anything, or make any marks below this line or outside borders

Section II extra writing space

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

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1 What values of x satisfy $3 - 2x \geq 8$?

A. $x \leq -\frac{11}{2}$

B. $x \geq -\frac{11}{2}$

☒ C. $x \leq -\frac{5}{2}$

D. $x \geq \frac{5}{2}$

$$3 - 2x \geq 8 - 3$$

$$-2x \geq 5$$

$$x \leq -\frac{5}{2}$$

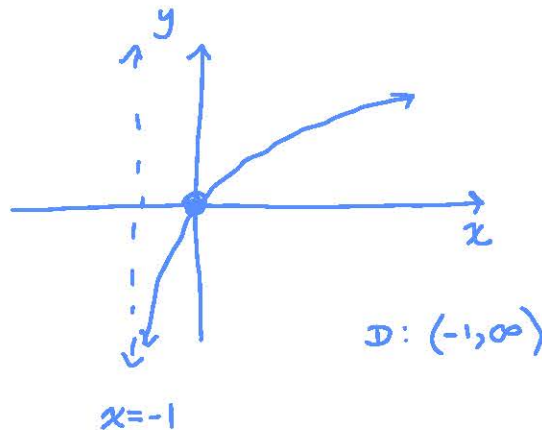
2 Which of the following represents the domain of the function $f(x) = \ln(x+1)$?

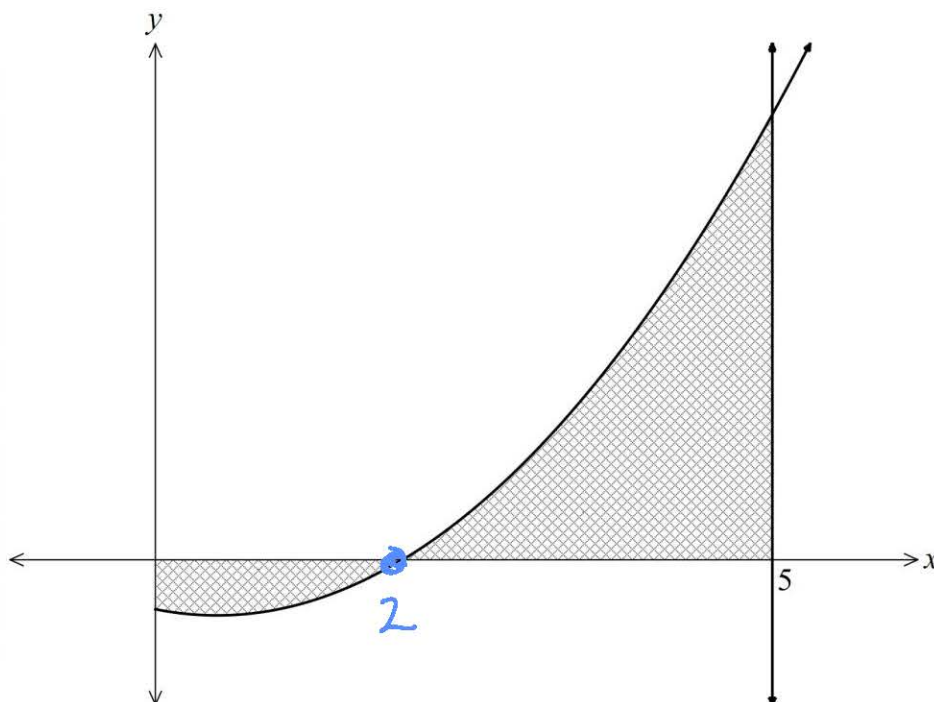
A. $[-\infty, -1)$

B. $(-\infty, 1)$

C. $[-1, \infty)$

☒ D. $(-1, \infty)$





- 3 The shaded region in the diagram above shows the area, A , bounded by the curve $y = x^2 - x - 2$, the x -axis and the lines $x = 0$ and $x = 5$?

Which expression will give the area, A , of the shaded region?

A. $A = \left| \int_0^1 (x^2 - x - 2) dx \right| + \int_1^5 (x^2 - x - 2) dx$

B. $A = \int_0^2 (x^2 - x - 2) dx + \left| \int_2^5 (x^2 - x - 2) dx \right|$

C. $A = \int_2^0 (x^2 - x - 2) dx + \int_2^5 (x^2 - x - 2) dx$

D. $A = \int_1^5 (x^2 - x - 2) dx$

$$\begin{aligned} x^2 - x - 2 &= 0 \\ (x-2)(x+1) &= 0 \\ x &= 2, x = -1. \end{aligned}$$

$$\left| \int_0^2 (x^2 - x - 2) dx \right| + \int_2^5 (x^2 - x - 2) dx$$

- 4 Which of the following represent respectively the radius and coordinates of the centre of the circle $x^2 - 6x + y^2 + 8y + 21 = 0$?

A. Radius of 2 **and** centre $(-3, 4)$

☒ B. Radius of 2 **and** $(3, -4)$

C. Radius of 4 **and** $(-3, 4)$

D. Radius of 4 **and** $(3, -4)$

$$(x-3)^2 - 9 + (y+4)^2 - 16 + 21 = 0$$

$$(x-3)^2 + (y+4)^2 = 4$$

centre: $(3, -4)$ radius = 2

- 5 What is the period of $y = \tan 3x$?

☒ A. $\frac{\pi}{3}$

B. $\frac{\pi}{6}$

C. $\frac{2\pi}{3}$

D. 3π

$$\text{period} = \frac{\pi}{b} = \frac{\pi}{3}$$

- 6 The values of x for which the geometric series $2 + 6x + 18x^2 + \dots$ has a limiting sum are:

A. $x < \frac{1}{3}$

B. $x \leq \frac{1}{3}$

☒ C. $-\frac{1}{3} < x < \frac{1}{3}$

D. $-\frac{1}{3} \leq x \leq \frac{1}{3}$

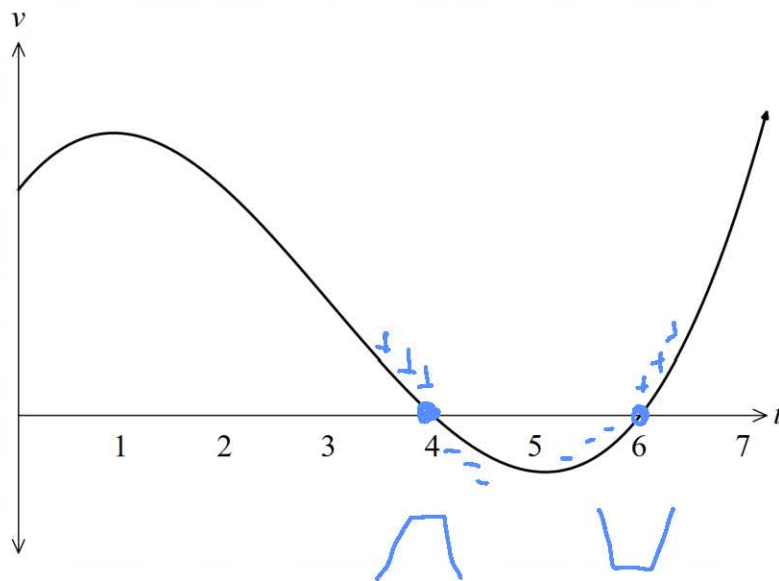
$$-1 < r < 1$$

$$r = \frac{6x}{2} = 3x$$

$$-1 < 3x < 1$$

$$-\frac{1}{3} < x < \frac{1}{3}$$

- 7 A particle is moving along a straight line. The graph below shows the velocity, v , of the particle at any time t seconds.



For what value of t is the displacement, x , a maximum?

- A. 1
- ☒ B. 4
- C. 5
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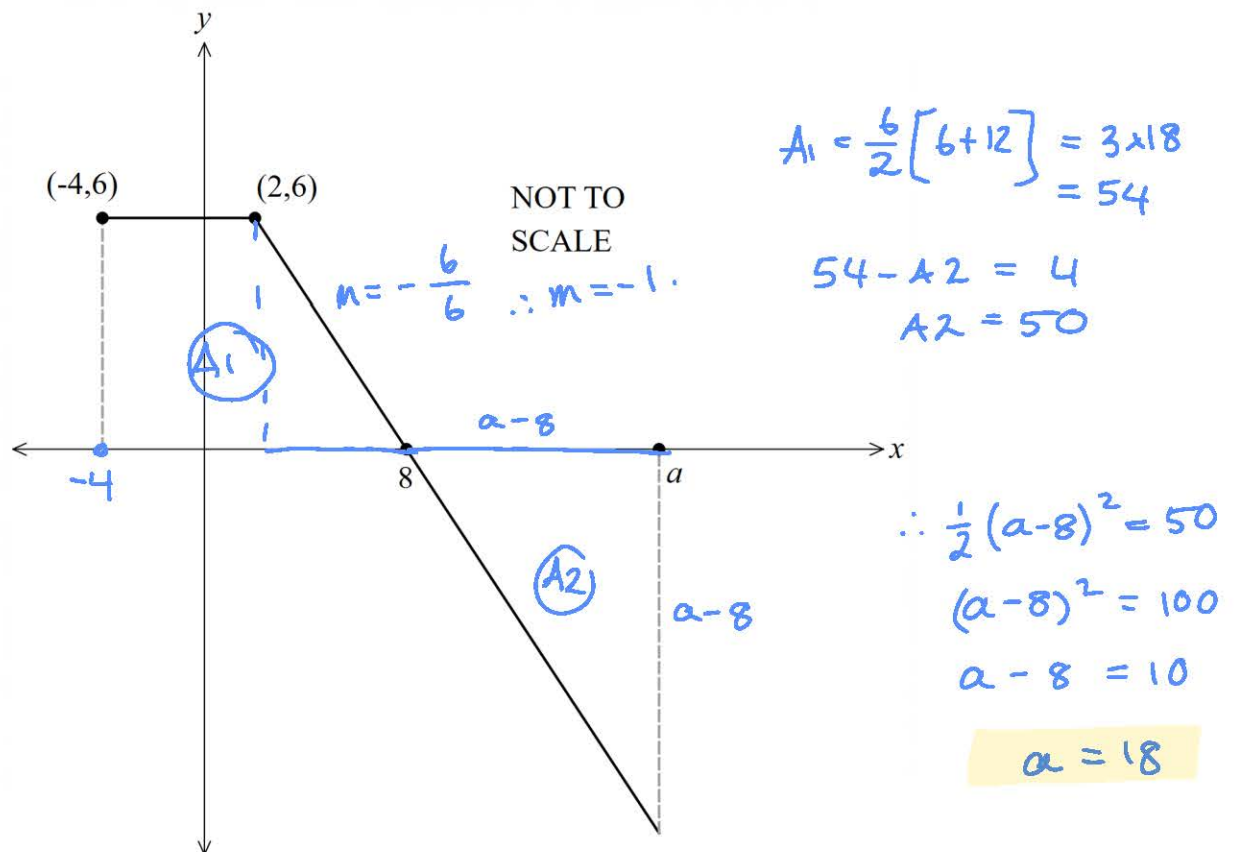
x	1	2	3	4	5
$P(X=x)$	a	$3a$	$5a$	$2a$	a

What is the mean, $E(X)$, for this probability distribution?

- A. $\frac{1}{12}$
- B. $1\frac{3}{12}$
- ☒ C. $2\frac{11}{12}$
- D. 3

$$\begin{aligned}
 a + 3a + 5a + 2a + a &= 1 \\
 12a &= 1 \\
 a &= \frac{1}{12} \\
 \therefore E(X) &= a + 6a + 15a + 8a + 5a \\
 &= 35a \\
 &= 35 \times \frac{1}{12} = \frac{35}{12} = 2\frac{11}{12}
 \end{aligned}$$

- 9 The graph of $y = f(x)$ is a piecewise function with a domain of $[-4, a]$, where $a > 8$.

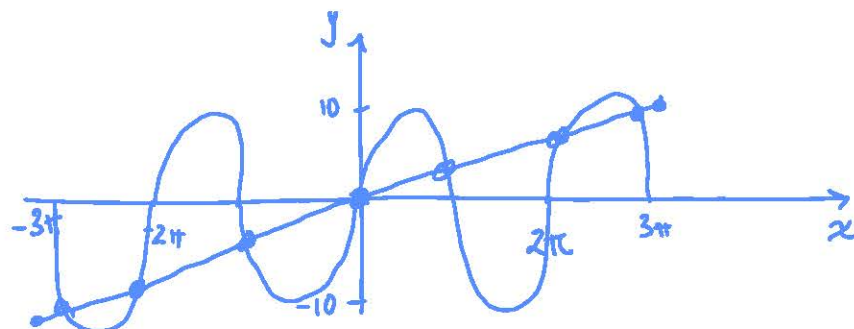


Find the value of a given that $\int_{-4}^a f(x) \, dx = 4$.

- A. 16
- B. 18**
- C. 20
- D. 22

- 10 How many solutions does the equation $x = 10 \sin x$ have, given $x \in [-3\pi, 3\pi]$?

- A. 4
- B. 5
- C. 6
- D. 7**





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Question 11 (4 marks)

The first three terms of an arithmetic sequence are 4, 9 and 14.

- (a) Find the 20th term.

2

$$T_{20} = 4 + (20-1) \times 5 \quad 1 \text{mk}$$

$$T_{20} = 99 \quad 1 \text{mk}$$

[2mk all correct
1mk using n^{th} term rule]

- (b) Find the sum of the first 20 terms.

2

$$S_{20} = \frac{20}{2} [4 + 99] \quad 1 \text{mk}$$

$$= 1030 \quad 1 \text{mk}$$

[2mk all correct
1mk using S_n Rule]

$$\text{Alternate: } S_{20} = \frac{20}{2} [2 \times 4 + (20-1) \times 5]$$

$$S_{20} = 1030$$

Question 12 (4 marks)

Find the value(s) of x such that $25^{x^2-2x} = \left(\frac{1}{125}\right)^{4x+2}$.

4

$$(5^2)^{x^2-2x} = (5^{-3})^{4x+2}$$

1mk

$$5^{2x^2-4x} = 5^{-12x-6}$$

$$\therefore 2x^2 - 4x = -12x - 6 \quad 1mk$$

$$2x^2 + 8x + 6 = 0$$

$$x^2 + 4x + 3 = 0$$

$$(x+3)(x+1) = 0$$

$$x = -3, x = -1 \quad 1mk \text{ each}$$

4mks all correct

3mks one correct solution

2mks equating powers

1mk attempt common base

Question 13 (3 marks)

Find the equation of the tangent to the curve $y = \frac{1}{\sqrt{x}}$ at the point $\left(4, \frac{1}{2}\right)$.

Give your final answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.

3

$$y = x^{-\frac{1}{2}}$$

$$y' = -\frac{1}{2}x^{-\frac{3}{2}} \quad 1mk$$

$$\therefore x = 4, m = -\frac{1}{16} \quad 1mk \quad @ (4, \frac{1}{2})$$

$$\therefore y - \frac{1}{2} = -\frac{1}{16}(x - 4)$$

$$16y - 8 = -x + 4$$

$$x + 16y - 12 = 0 \quad 1mk$$

3mks all correct

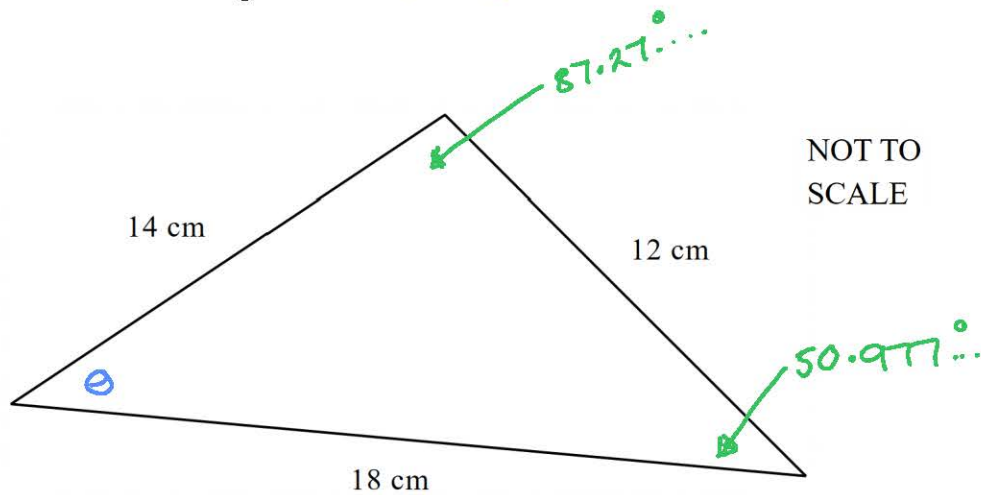
2mks correct gradient

1mk derivative

Question 14 (3 marks)

Find the area of the triangle below. *to 1 dp.*

3



$$\cos \theta = \frac{14^2 + 18^2 - 12^2}{2 \times 14 \times 18} \quad 1 \text{mk}$$

$$\theta = 41.7522052... \quad 1 \text{mk}$$

3mks all correct
2mks a correct angle
1mk cosine attempt

$$\therefore A = \frac{1}{2} \times 14 \times 18 \times \sin 41.7522052...$$

$$A = 83.9 \text{ cm}^2 \quad 1 \text{mk}$$

Question 15 (2 marks)

Find $\int \frac{5x}{x^2+1} dx$.

2

$$= 5 \int \frac{x}{x^2+1} dx$$

$$= \frac{5}{2} \ln(x^2+1) dx$$

2mks all correct
1mk $\ln(x^2+1)$

Question 16 (5 marks)

Differentiate each of the following expressions with respect to x , leaving your answer in simplest form where possible:

(a) $(2x^3 - 1)^5$

1

$$\frac{d}{dx} = 5(2x^3 - 1)^4 \cdot 6x^2$$
$$= 30x^2(2x^3 - 1)^4$$

[1mk correct]

(b) $x \cos x$

2

$$\frac{d}{dx} = v u' + u v'$$

$$u = x \quad v = \cos x \quad \text{1mk}$$

$$= \cos x \cdot 1 + x \cdot (-\sin x)$$

$$u' = 1 \quad v' = -\sin x$$

$$= \cos x - x \sin x$$

1mk

[2mks all correct
1mk correct derivatives]

(c) $\frac{2x}{\ln x}$

2

$$\frac{d}{dx} = \frac{v u' - u v'}{v^2}$$

$$u = 2x \quad v = \ln x \quad \text{1mk}$$

$$= \frac{\ln x \cdot 2 - 2x \cdot \frac{1}{x}}{(\ln x)^2}$$

$$u' = 2 \quad v' = \frac{1}{x}$$

$$= \frac{2 \ln x - 2}{2 \ln x}$$

[2mks all correct
1mk correct derivatives]

$$= \frac{\ln x - 1}{\ln x} \quad \text{1mk}$$

$$0 \leq 2\theta \leq 2\pi$$

Question 17 (2 marks)

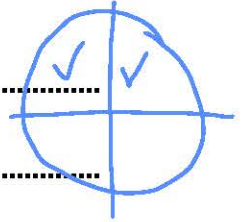
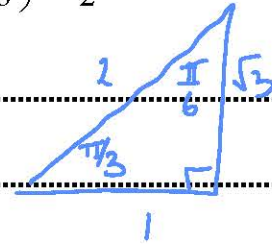
$$-\frac{\pi}{3} \leq 2\theta - \frac{\pi}{3} \leq \frac{5\pi}{3}$$

Find all the values of θ , where $0 \leq \theta \leq \pi$, such that $\sin\left(2\theta - \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

2

$$2\theta - \frac{\pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}$$

$$2\theta = \frac{2\pi}{3}, \pi$$



$$\therefore \theta = \frac{\pi}{3}, \frac{\pi}{2}$$

1mk

2mks all correct
1mk $\frac{\pi}{3}$ seen.

Question 18 (3 marks)

A and B are independent events with $P(A) = 0.28$ and $P(A \cup B) = 0.75$.

Find $P(B)$.

3

$$P(A \cap B) = P(A) \times P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

1mk

$$\therefore 0.75 = 0.28 + P(B) - 0.28P(B)$$

1mk

$$0.47 = 0.72P(B)$$

$$\therefore P(B) = \frac{47}{72}$$

1mk

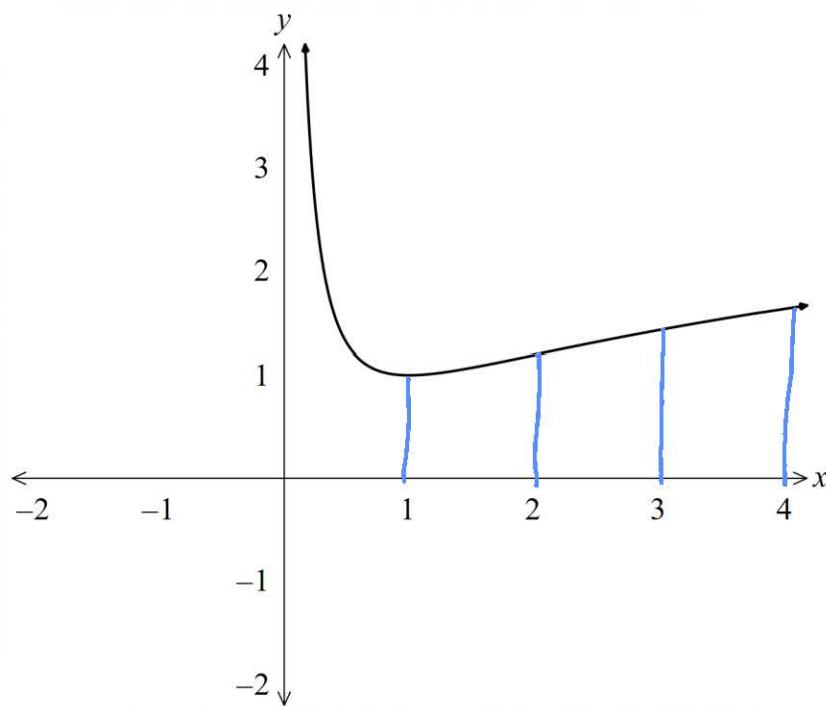
3mks all correct

2mks correct substitution

1mk either formula seen

Question 19 (2 marks)

The graph of $y = \frac{1}{x} + \ln(x)$ is shown over part of its domain.



Use **three** applications of the trapezoidal rule to find the approximate area between the curve, the x -axis and the lines $x=1$ and $x=4$. Give your answer correct to 2 decimal places.

2

$$A \doteq \frac{1}{2} \left[1 + \ln 1 + \frac{1}{4} + \ln 4 + 2 \left(\frac{1}{2} + \ln 2 + \frac{1}{3} + \ln 3 \right) \right]$$

$$A \doteq 3.94 \text{ u}^2$$

2 mks all correct

1 mk substitution into correct formula

Question 20 (5 marks)

A table of future value interest factors for an annuity of \$1000 is shown.

Rate Period	Interest rate per period				
	1%	1.5%	3%	4.5%	6%
3	3030.10	3045.23	3090.90	3137.03	3183.60
6	6152.02	6229.55	6468.41	6716.89	6975.32
12	12682.50	13041.21	14192.03	15464.03	16869.94
24	26973.46	28633.52	34426.47	41689.20	50815.58

- (a) Taliah is saving for a new car. She considers contributing \$7 500 each year for the next 6 years.

If the interest rate is 6% per annum, compounding annually, how much money will Taliah have saved after 6 years?

2

$$FV = PV \times IF$$

$$FV = 7.5 \times 6975.32$$

$$FV = \$52314.90$$

2mk all correct

1mk for Interest Factor

- (b) Taliah requires \$85 000 to buy her dream car.

She decides to make regular payments into an account earning 6% per annum, compounding **quarterly** for the next 6 years.

Given that Taliah will make her contribution every 3 months, how much more will she pay per year, than in part (a)?

3

$$FV = PV \times IF$$

$$85000 = PV \times 28633.52$$

$$PV = 2.9685 \dots$$

$$PV = \$2968.55 \times 4$$

$$= \$11874.195 - 7500$$

$$= \$4374.19 \text{ more each year}$$

3mk all correct

2mk \$2968.55

1mk Interest Factor

Question 21 (3 marks)

Solve the equation $2\log_5(4) - \log_5(2) + \log_5(x) = 2$ for x .

3

$$\log_5 16 - \log_5 2 + \log_5 x = 2 \quad 1\text{mk}$$

$$\log_5(8x) = 2\log_5 5$$

$$\log_5 8x = \log_5 25 \quad 1\text{mk}$$

$$\therefore 8x = 25$$

$$x = \frac{25}{8}$$

1mk

3mk all correct

2mk equating sides same base

1mk any correct log conversion

Question 22 (3 marks)

Consider the function f such that $f'(x) = \frac{8x}{x^2+4}$. The graph of $y = f(x)$ passes through the point $(2, 9\ln 2)$.

Find $f(x)$.

3

$$f(x) = \int \frac{8x}{x^2+4} dx$$

$$= 8 \int \frac{x}{x^2+4} dx$$

$$f(x) = 4 \ln(x^2+4) + C$$

$$9\ln 2 = 4 \ln(8) + C$$

$$C = 9\ln 2 - 4 \ln 2^3$$

$$C = 9\ln 2 - 12 \ln 2$$

$$C = -3 \ln 2$$

$$\therefore f(x) = 4 \ln(x^2+4) - 3 \ln 2$$

3mks all correct
3mks $4 \ln(x^2+4) + C$
1mk $\ln(x^2+4)$ seen

Question 23 (5 marks)

Let $f(x) = xe^x$ and $g(x) = 4x - x^2$.

Solve $(f \circ g)(x) = 0$.

5

$$(f \circ g)(x) = (4x - x^2) \cdot e^{4x - x^2}$$

$$0 = (4x - x^2) e^{4x - x^2}$$

$$\therefore 4x - x^2 = 0 \quad e^{4x - x^2} \neq 0 \therefore \text{No solution}$$

$$x(4 - x) = 0$$

$$x = 0, x = 4.$$

5mks all correct

4mks $x=0, x=4$ and no exclusion

3mks correct composite $= 0$

2mks correct composite

1mk any correct substitution

Please turn over

Question 24 (9 marks)

The curve $y = x^3 + ax^2 + 7x$ has a stationary point at $x = 1$.

- (a) Find the value of a .

2

$$y' = 3x^2 + 2ax + 7 \text{ 1mk}$$

$$y' = 0 \text{ for stat pt.}$$

$$0 = 3(1)^2 + 2a(1) + 7$$

$$0 = 3 + 2a + 7$$

$$2a = -10 \quad \therefore a = -5 \text{ 1mk}$$

2mks all correct
1mk correct derivative

- (b) Find the coordinates of all stationary points **and** determine their nature.

3

$$y = x^3 - 5x^2 + 7x$$

$$y' = 3x^2 - 10x + 7$$

$$0 = 3x^2 - 10x + 7$$

$$y'' = 6x - 10$$

$$x = \frac{7}{3}, y'' = 4 \therefore y'' > 0 \rightarrow \text{min}$$

$$x = 1, y'' = -4 \therefore y'' < 0 \rightarrow \text{max}$$

$$\frac{p}{q} \mid \frac{r}{s} \quad (-7, -3) \therefore 0 = 3x^2 - 7x - 3x + 7$$

$$0 = x(3x - 7) - 1(3x - 7)$$

$$0 = (3x - 7)(x - 1)$$

$$x = \frac{7}{3}, \quad x = 1$$

$$y = \frac{49}{27}, \quad y = 3$$

$$\therefore \left(\frac{7}{3}, \frac{49}{27}\right) \text{ minimum}$$

and $(1, 3) \text{ maximum}$

3mks all correct

2mks coordinates, no nature

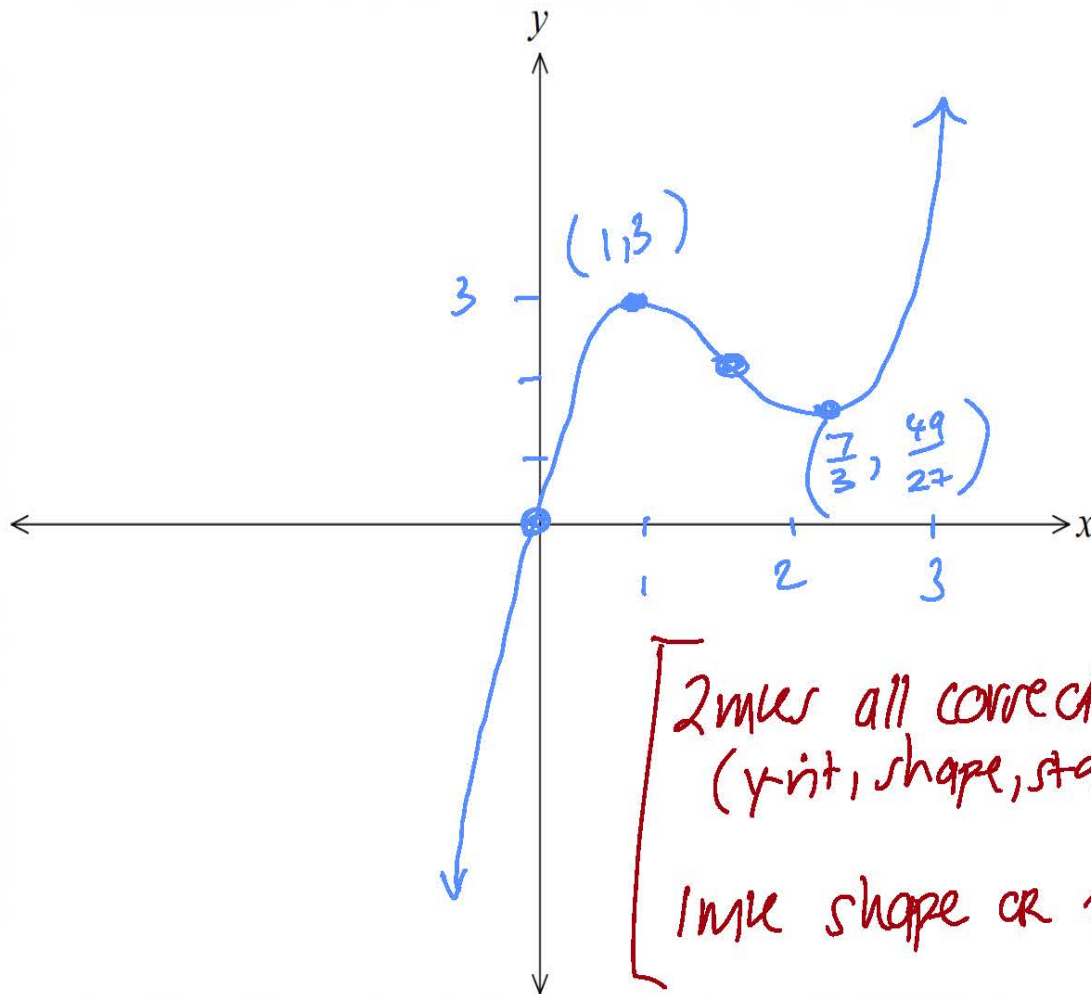
1mk letting $y' = 0$ correctly

Question 24 continues on page 19

Question 24 (continued)

- (c) Sketch the graph of $y = f(x)$ showing any stationary points and its y -intercept.

2



- (d) Find any values of x such that the curve is concave up.

2

$$y'' = 6x - 10$$

$$0 = 6x - 10$$

$$6x = 10$$

$$x = \frac{5}{3}$$

$$x > \frac{5}{3}$$

[2mks all correct]
1mk $y'' = 0$

End of Question 24

Please turn over

Question 25 (3 marks)

(a) Find $\frac{d}{dx}(e^{\tan x})$.

1

$$\frac{d}{dx} = \sec^2 x e^{\tan x}$$

[1mk only]

(b) Hence, or otherwise, find $\int \frac{2e^{\tan x}}{3-3\sin^2 x} dx$.

2

$$\int \frac{2e^{\tan x}}{3(1-\sin^2 x)} dx$$

$$= \frac{1}{3} \int \frac{2e^{\tan x}}{\cos^2 x} dx$$

$$= \frac{2}{3} \int \sec^2 x \cdot e^{\tan x} dx$$

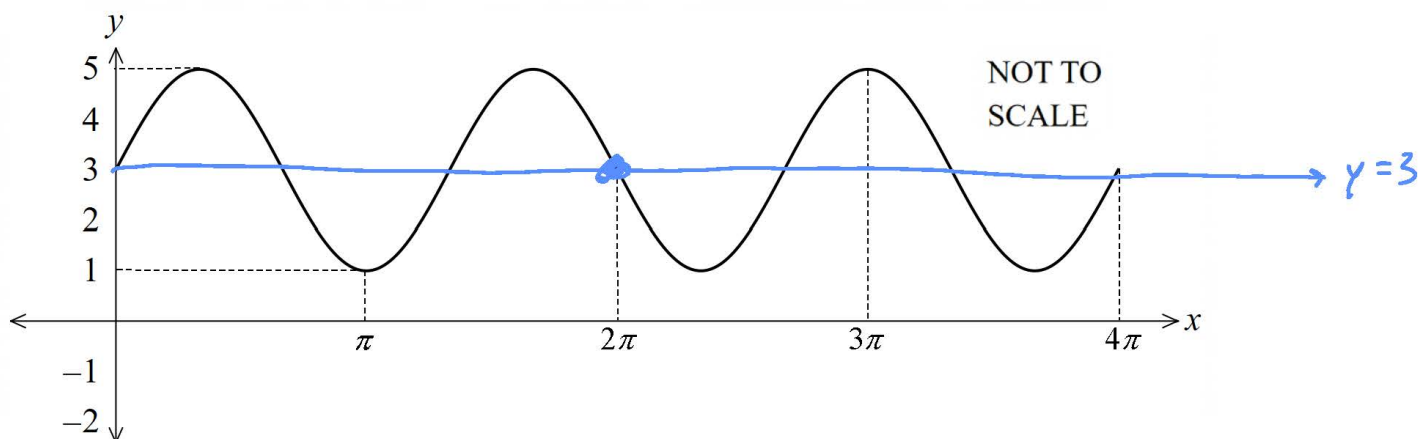
2mks all correct
1mk working towards

1mk

$$= \frac{2}{3} e^{\tan x} + C \quad \text{or} \quad \frac{2e^{\tan x}}{3} + C \quad \text{1mk.}$$

Question 26 (4 marks)

A trigonometric function $y = f(x)$ is drawn below in the domain $[0, 4\pi]$.



- (a) State the amplitude of $y = f(x)$.

1

amplitude = 2 [1mk only]

- (b) Determine the equation of the function $y = f(x)$.

Express your final answer in the form $y = a \sin(bx) + d$.

3

$y = 2 \sin\left(\frac{3}{2}x\right) + 3$

3mks all correct
2mks 2 correct variables
1mk 1 correct variable

Question 27 (7 marks)

$$4 + 8x$$

The following table shows values of x and y .

x	4	13	26	30	35	50
y	1.8	2.9	3.8	4.2	5.7	5.1

- (a) Calculate Pearson's correlation coefficient for the data represented in the table above, correct to 2 decimal places.

1

0.91

[1mk only]

- (b) Identify the direction **and** strength of the linear association between x and y .

2

strong positive correlation.
1mk 1mk

[2mks all correct
1mk either]

- (c) State the equation of the least-squares regression line in the form $y = ax + b$.

2

$$y = 0.08x + 1.82$$

1mk

1mk.

[2mks all correct
1mk either a or b]

- (d) Use the regression equation from part (c) to estimate the value of x when $y = 4.8$ to 2 decimal places.

2

$$4.8 = 0.08x + 1.82$$

1mk

[2mks all correct
1mk substitution]

$$\therefore x = 37.25$$

1mk

Question 28 (5 marks)

A continuous random variable X has a function given by

$$f(x) = \begin{cases} \pi \sin(2\pi x) & \text{if } 0 \leq x \leq \frac{1}{2} \\ 0 & \text{elsewhere} \end{cases}$$

- (a) Show that the function $f(x)$ is a probability density function.

3

$$\begin{aligned} \therefore \int_0^{\frac{1}{2}} \pi \sin(2\pi x) dx &= 1 \\ -\pi \left[\frac{\cos(2\pi x)}{2\pi} \right]_0^{\frac{1}{2}} &= 1 \\ -\frac{1}{2} [\cos(2\pi x)]_0^{\frac{1}{2}} &= 1 \\ -\frac{1}{2} [\cos \pi - \cos 0] &= 1 \\ -\frac{1}{2} [-1 - 1] &= 1 \\ 1 &= 1 \quad \therefore \text{LHS} = \text{RHS} \quad \text{it is a PDF} \end{aligned}$$

*3mks all correct
2mks correct integral
1mk let integral = 1*

- (b) Find the mode of X .

2

Mode: $f'(x) = 2\pi \cdot \pi \cos(2\pi x)$

Let $f'(x) = 0$ for turning point -

$$0 = 2\pi^2 \cos(2\pi x)$$

$$0 = \cos(2\pi x)$$

$$(2\pi x) = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$x = \frac{1}{4}, \frac{3}{4}$$

3.14 -3.14

2mks all correct

*1mk $x = \frac{1}{4}, x = \frac{3}{4}$
with no test*

$$f''(x) = -4\pi^3 \sin(2\pi x)$$

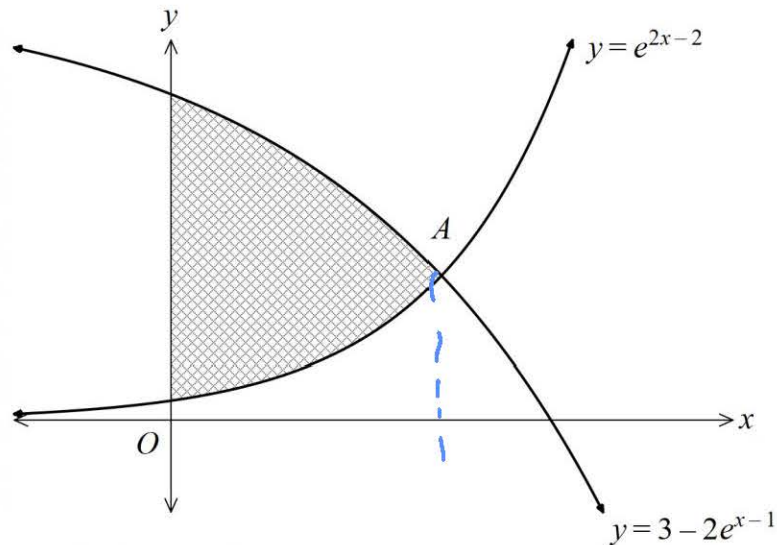
$x = \frac{1}{4}, f''(x) = -12\pi$ *max*

$x = \frac{3}{4}, f''(x) = 12\pi$ *min.*

$\therefore x = \frac{1}{4}$

Question 29 (4 marks)

The shaded region is bound by the functions $y = e^{2x-2}$, $y = 3 - 2e^{x-1}$ and the y -axis, as shown in the diagram. Point A is the intersection of the two functions.



- (a) Show that the x -coordinate of point A is $x=1$.

2

Method 1: $e^{2x-2} = 3 - 2e^{x-1}$

$$(e^{x-1})^2 + 2e^{x-1} - 3 = 0$$

$$u^2 + 2u - 3 = 0$$

$$(u+3)(u-1) = 0$$

$$u = -3, u = 1$$

$$\therefore e^{x-1} = -3 \quad e^{x-1} = 1$$

$$\text{No solution} \quad x-1 = 0$$

$$x = 1$$

Method 2:

$$e^{2(1)-2} = e^0 = 1$$

$$3 - 2e^{1-1} = 3 - 2 = 1$$

$$\therefore \text{LHS} = \text{RHS} = 1$$

[Implies all correct
Imu working towards]

Question 29 continues on page 25

Question 29 (continued)

(b) Find the **exact** area of the shaded region.

2

$$A = \int_0^1 (3 - 2e^{x-1} - e^{2x-2}) dx$$

$$A = \left[3x - 2e^{x-1} - \frac{e^{2x-2}}{2} \right]_0^1$$

$$A = \left[3 - 2 - \frac{1}{2} \right] - \left[0 - 2e^{-1} - \frac{e^{-2}}{2} \right]$$

$$A = \frac{1}{2} - \left[-\frac{2}{e} - \frac{1}{2e^2} \right]$$

$$A = \frac{1}{2} + \frac{2}{e} + \frac{1}{2e^2} \quad \text{or} \quad A = \frac{e^2 + 4e + 1}{2e^2} \quad \text{a}^2$$

[2mk all correct
1mk correct integral]

End of Question 29

Please turn over

Question 30 (8 marks)

Kareem borrows \$140 000 to start his own business. He plans to repay the loan in equal monthly instalments of \$ M at 6% per annum reducible interest. Interest is calculated and charged just before each repayment is made.

Let A_n be the amount owing after n repayments (in dollars).

- (a) Show that $A_n = 140\,000(1.005)^n - 200M(1.005^n - 1)$.

3

$$A_1 = 140,000(1.005) - m$$

$$A_2 = [140,000(1.005) - m] \times 1.005 - m$$

$$A_2 = 140,000(1.005)^2 - m(1.005) - m$$

$$A_3 = [140,000(1.005)^2 - m(1.005) - m] \times 1.005 - m$$

$$A_3 = 140,000(1.005)^3 - m(1.005)^2 - m(1.005) - m$$

$$A_3 = 140,000(1.005)^3 - m(1 + 1.005 + 1.005^2)$$

$$A_n = 140,000(1.005)^n - m(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$$

$$A_n = 140,000(1.005)^n - m \times \frac{(1.005^n - 1)}{0.005}$$

$$A_n = 140,000(1.005)^n - 200m(1.005^n - 1)$$

3mks all correct
2mks for A2/A3
1mk for A1

Question 30 continues on page 27

Question 30 (continued)

- (b) Kareem would like to repay the loan in exactly 15 years. Calculate the amount of each instalment.

2

$$0 = 140,000(1.005)^{180} - 200m(1.005^{180} - 1) \quad 1mk$$

$$200m(1.005^{180} - 1) = 140,000(1.005)^{180}$$

$$m = \frac{140,000(1.005)^{180}}{200(1.005^{180} - 1)}$$

$$m = \$1181.40$$

1mk

[2 mks all correct
1mk correct substitution]

- (c) Kareem can afford to repay \$1 850 per month. How many years would it take him to repay the loan?

3

$$0 = 140,000(1.005)^n - 200(1850)(1.005^n - 1) \quad 1mk$$

$$0 = 140,000(1.005)^n - 370,000(1.005^n) + 379,000$$

$$-370,000 = (1.005)^n(-230,000)$$

$$1.005^n = \frac{370,000}{230,000} \quad 1mk$$

$$\therefore n = \frac{\ln\left(\frac{37}{23}\right)}{\ln 1.005}$$

$$\therefore n = 95.3222 \dots$$

$$\therefore n = 96 \text{ payments}$$

1mk

$$\therefore 7.94 \text{ years} / 8 \text{ years}$$

[3mks all correct
2mks correct exponential
1mk correct substitution]

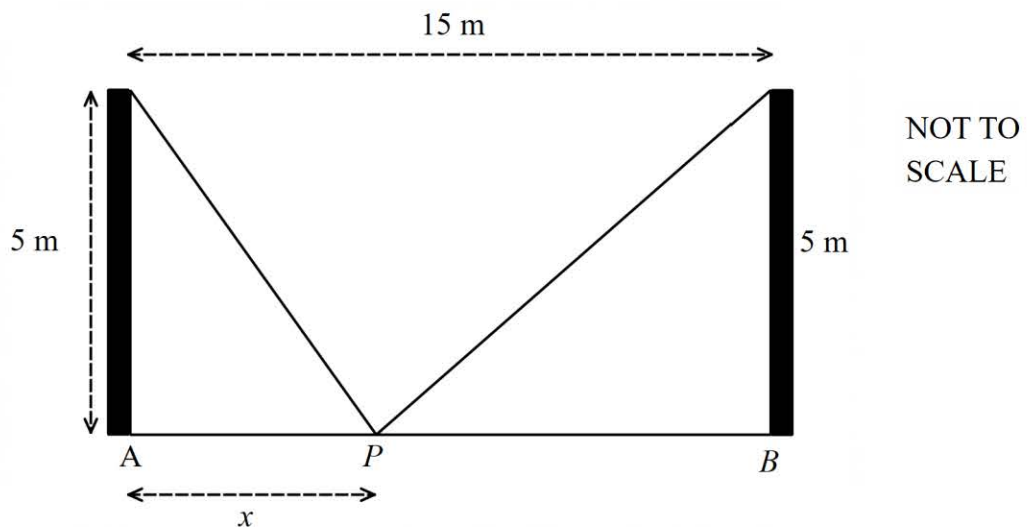
End of Question 30

Please turn over

Question 31 (6 marks)

The diagram shows two 5 m poles, A and B , that are 15 metres apart. The ends of a single length of wire are attached to the top of each pole. The length of wire is then attached to a stake in the ground between them at a moveable point, P , as shown in the diagram below.

Let x be the distance between A and P .



- (a) Show that the length of the wire, L , is given by $L = \sqrt{x^2 + 25} + \sqrt{x^2 - 30x + 250}$. 1

$$L = \sqrt{x^2 + 5^2} + \sqrt{(15-x)^2 + 5^2}$$

$$L = \sqrt{x^2 + 25} + \sqrt{225 - 30x + x^2 + 25}$$

$$L = \sqrt{x^2 + 25} + \sqrt{x^2 - 30x + 250}$$

[mk only]

Question 31 continues on page 29

Question 31 (continued)

(b) Show that $\frac{dL}{dx} = \frac{x}{\sqrt{x^2+25}} + \frac{x-15}{\sqrt{x^2-30x+250}}$.

2

$$L = (x^2+25)^{1/2} + (x^2-30x+250)^{1/2} \quad \text{1mk}$$

$$\frac{dL}{dx} = \frac{1}{2}(x^2+25)^{-1/2} \cdot 2x + \frac{1}{2}(x^2-30x+250)^{-1/2} \cdot (2x-30) \quad \text{1mk}$$

$$\frac{dL}{dx} = \frac{x}{\sqrt{x^2+25}} + \frac{x-15}{\sqrt{x^2-30x+250}}$$

(c) Find the value of x needed to minimise the length of wire, L used.

Justify your answer.

3

$$\text{Let } \frac{dL}{dx} = 0$$

$$\therefore 0 = \frac{x}{\sqrt{x^2+25}} + \frac{x-15}{\sqrt{x^2-30x+250}} \quad \textcircled{1}$$

$$0 = x\sqrt{x^2-30x+250} + (x-15)(\sqrt{x^2+25}) \quad \text{1mk } \frac{dL}{dx} = 0$$

$$-(x-15)(\sqrt{x^2+25}) = x\sqrt{x^2-30x+250}$$

$$(15-x)(\sqrt{x^2+25}) = x\sqrt{x^2-30x+250}$$

$$(15-x)^2(x^2+25) = x^2(x^2-30x+250)$$

$$(225-30x+x^2)(x^2+25) = x^4-30x^3+250x^2$$

$$225x^2-30x^3+x^4+5625-750x+25x^2-x^4+30x^3-250x^2=0$$

$$\therefore 750x = 5625$$

$$x = 7.5 \quad \textcircled{1}$$

$\therefore$ minimum

End of Question 31
End of paper

Test:

x	7	7.5	8
y'	-0.03	0	0.03

- 0 +

⌈

Ⓛ